Def. Let G be a group. If for some $x \in G$,

$$G = \{x^n \mid n \in \mathbb{Z}\} = \langle x \rangle$$

Then, $G = \langle x \rangle$ is called the cyclic group generated by $x \in G$.

Prop. Every subgroup S of a cyclic group $\langle x \rangle$ is cyclic.

Proof. If $S = \{1\} = \{x^0\}$, then it is a cyclic generated by 1.
If $S \neq \{1\}$, let $m \in \mathbb{Z}^+$ be the smallest number s.t. $x^m \in S$.
Cleary, $\langle x^m \rangle \subseteq S$. Conversely, let $x^k \in S \subseteq \langle x \rangle$ be given. Then,

$$k = mq + r \text{ for some } q, r \in \mathbb{Z} \text{ with } 0 \leq r < m.$$

So, $x^k = x^{mq+r} = x^{mq} \cdot x^r \in S$ and $x^{mq} \in S$ implies $x^{-mq} \in S$, so

$$x^{-mq} \cdot x^{mq} \cdot x^r = x^r \in S.$$

By the minimality, $r = 0$. So, $x^k = x^{mq} \in \langle x^m \rangle$. Thus $S \subseteq \langle x^m \rangle$. ▨

Prop. Let $G = \langle x \rangle$ be a group of order n. Then,

$$x^k \text{ generates subgroup of order of } n/\gcd(n,k).$$

Proof. Let $d = \gcd(n,k)$. Then, $n = dx$ and $k = dy$ for some $x, y \in \mathbb{N}$, with $\gcd(x,y) = 1$.
observe that $(a^k)^x = a^{kx} = a^{dxy} = a^{ny} = 1^y = 1$. So, $x = n/d$ is the multiple of order of x^k.

Now, let $m \in \mathbb{N}$ be the order of x^k. That is, the smallest positive integer s.t. $x^{km} = 1$.
Hence, n divides km, it equals to $dx \mid dym \Rightarrow x \mid ym$.
Since $\gcd(x,y) = 1$, x divides m, so $x = n/d \leq m$. ▨